How to set up monitoring

Prerequisites:

1. Running cluster (any cluster)
2. Install helm

Step 1: Set Up Helm (If Not Installed)

curl -fsSL -o get_helm.sh https://raw.githubusercontent.com/helm/helm/main/scripts/get-helm-3

chmod 700 get_helm.sh

./get_helm.sh

helm version

```
root@ip-172-31-37-250:/home/ubuntu# curl -fsSL -o get_helm.sh https://raw.githubusercontent.com/helm/helm/main/scripts/get-helm-3
root@ip-172-31-37-250:/home/ubuntu# chmod 700 get_helm.sh
root@ip-172-31-37-250:/home/ubuntu# ./get_helm.sh
Downloading https://get.helm.sh/helm-v3.17.2-linux-amd64.tar.gz
Verifying checksum... Done.
Preparing to install helm into /usr/local/bin
helm installed into /usr/local/bin/helm
root@ip-172-31-37-250:/home/ubuntu# helm version
version.BuildInfo{Version:"v3.17.2", GitCommit:"cc0bbbd6d6276b83880042c1ecb34087e84d41eb", GitTreeState:"clean", GoVersion:"go1.23.7"
}
root@ip-172-31-37-250:/home/ubuntu#
```

Step 2: Add the Prometheus & Grafana Helm Repository

helm repo add prometheus-community https://prometheus-community.github.io/helm-charts

helm repo update

```
root@ip-172-31-37-250:/home/ubuntu# helm repo add prometheus-community https://prometheus-community.github.io/helm-charts
helm repo update
"prometheus-community" has been added to your repositories
Hang tight while we grab the latest from your chart repositories...
...Successfully got an update from the "prometheus-community" chart repository
Update Complete. ⎈Happy Helming!⎈
root@ip-172-31-37-250:/home/ubuntu#
```

Step 3: Install Prometheus using Helm

kubectl create namespace monitoring

helm install prometheus prometheus-community/kube-prometheus-stack --namespace monitoring

```
root@ip-172-31-37-250:/home/ubuntu# kubectl create namespace monitoring
helm install prometheus prometheus-community/kube-prometheus-stack --namespace monitoring
namespace/monitoring created
NAME: prometheus
LAST DEPLOYED: Thu Apr  3 05:26:20 2025
NAMESPACE: monitoring
STATUS: deployed
REVISION: 1
NOTES:
kube-prometheus-stack has been installed. Check its status by running:
  kubectl --namespace monitoring get pods -l "release=prometheus"

Get Grafana 'admin' user password by running:

  kubectl --namespace monitoring get secrets prometheus-grafana -o jsonpath="{.data.admin-password}" | base64 -d ; echo

Access Grafana local instance:

  export POD_NAME=$(kubectl --namespace monitoring get pod -l "app.kubernetes.io/name=grafana,app.kubernetes.io/instance=prometheus" -oname)
  kubectl --namespace monitoring port-forward $POD_NAME 3000

Visit https://github.com/prometheus-operator/kube-prometheus for instructions on how to create & configure Alertmanager and Prometheus instances using the Operator.
root@ip-172-31-37-250:/home/ubuntu#
```

Step 4: Expose Grafana as a LoadBalancer

kubectl patch svc prometheus-grafana -n monitoring -p '{"spec": {"type": "LoadBalancer"}}'

```
root@ip-172-31-37-250:/home/ubuntu# kubectl patch svc prometheus-grafana -n monitoring -p '{"spec": {"type": "LoadBalancer"}}'
service/prometheus-grafana patched
root@ip-172-31-37-250:/home/ubuntu#
```

Step 5: Get the Grafana LoadBalancer URL

kubectl get svc -n monitoring prometheus-grafana

```
root@ip-172-31-37-250:/home/ubuntu# kubectl get svc -n monitoring prometheus-grafana
NAME                 TYPE           CLUSTER-IP     EXTERNAL-IP                                                              PORT(S)        AGE
prometheus-grafana   LoadBalancer   10.100.231.21  a7324618efd1e41bebbcc2e3cf29117e-1926344286.sa-east-1.elb.amazonaws.com  80:31556/TCP   5m11s
root@ip-172-31-37-250:/home/ubuntu#
```

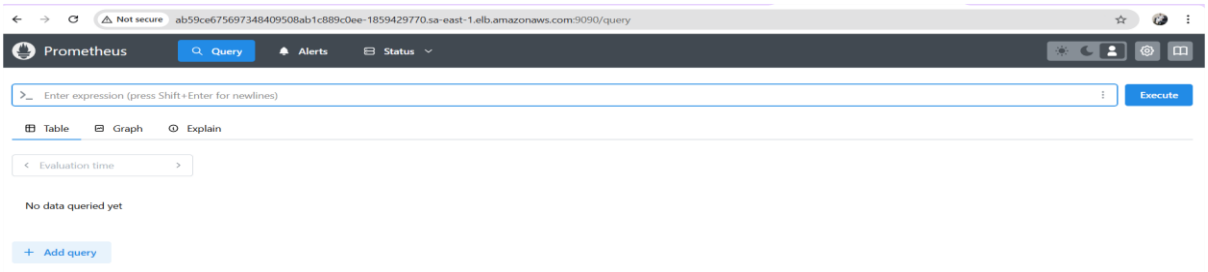
Expose Prometheus as a Load Balancer

Patch the Prometheus Service

kubectl patch svc prometheus-kube-prometheus-prometheus -n monitoring -p '{"spec": {"type": "LoadBalancer"}}'

```
root@ip-172-31-37-250:/home/ubuntu# kubectl get svc -n monitoring
NAME                                      TYPE           CLUSTER-IP       EXTERNAL-IP                                                                PORT(S)
                AGE
alertmanager-operated                     ClusterIP      None             <none>                                                                     9093/TCP,9094/TCP,9
094/UDP         23m
prometheus-grafana                        LoadBalancer   10.100.231.21    a7324618efd1e41bebbcc2e3cf29117e-1926344286.sa-east-1.elb.amazonaws.com    80:31556/TCP
                23m
prometheus-kube-prometheus-alertmanager   ClusterIP      10.100.183.196   <none>                                                                     9093/TCP,8080/TCP
                23m
prometheus-kube-prometheus-operator       ClusterIP      10.100.71.177    <none>                                                                     443/TCP
                23m
prometheus-kube-prometheus-prometheus     LoadBalancer   10.100.139.234   ab59ce675697348409508ab1c889c0ee-1859429770.sa-east-1.elb.amazonaws.com    9090:30744/TCP,8080
:30169/TCP      23m
prometheus-kube-state-metrics             ClusterIP      10.100.20.61     <none>                                                                     8080/TCP
                23m
prometheus-operated                       ClusterIP      None             <none>                                                                     9090/TCP
                23m
prometheus-prometheus-node-exporter       ClusterIP      10.100.100.98    <none>                                                                     9100/TCP
                23m
root@ip-172-31-37-250:/home/ubuntu#
```

Access Prometheus on web browser

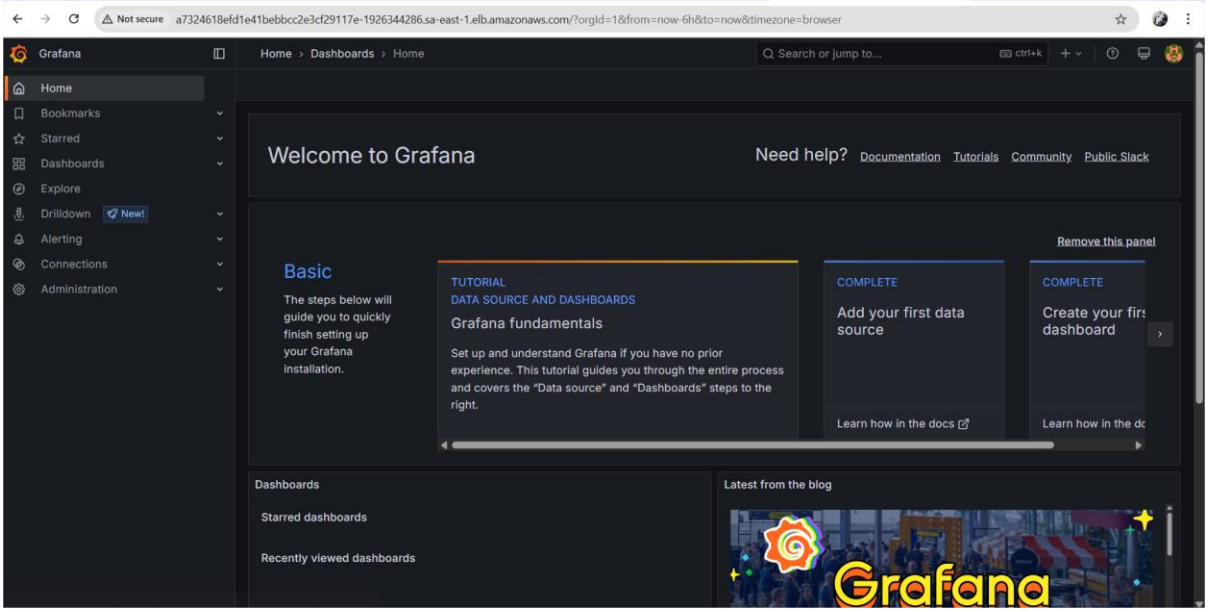


Step 6: Access Grafana

Open **http://<EXTERNAL-IP>:80** in your browser.

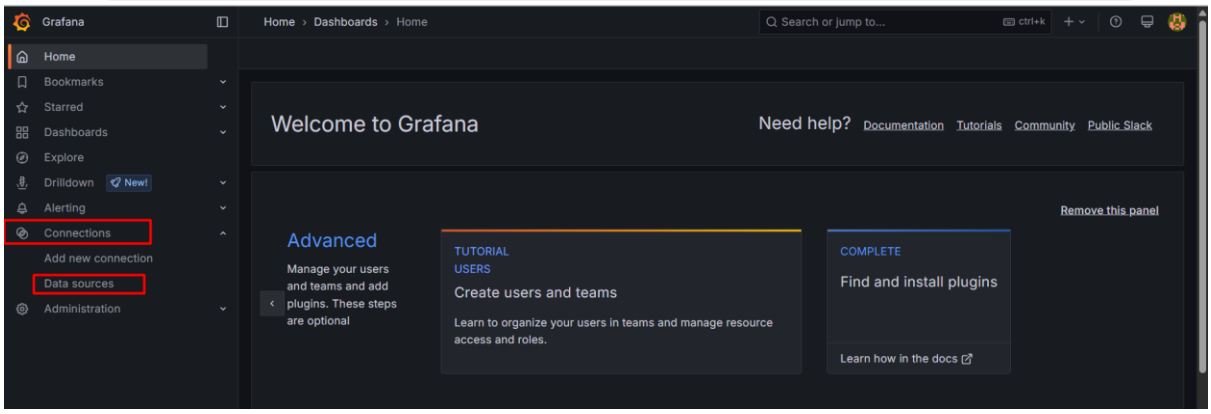
Default Login Credentials:

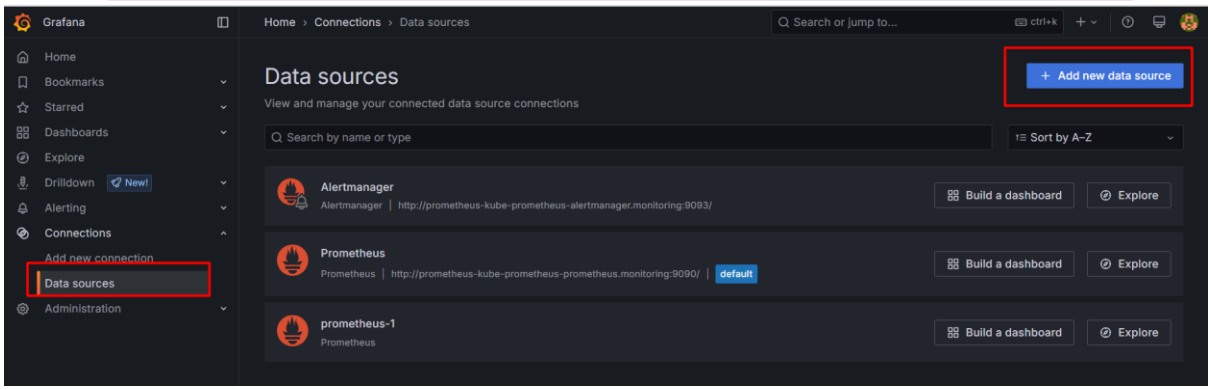
- **Username:** admin



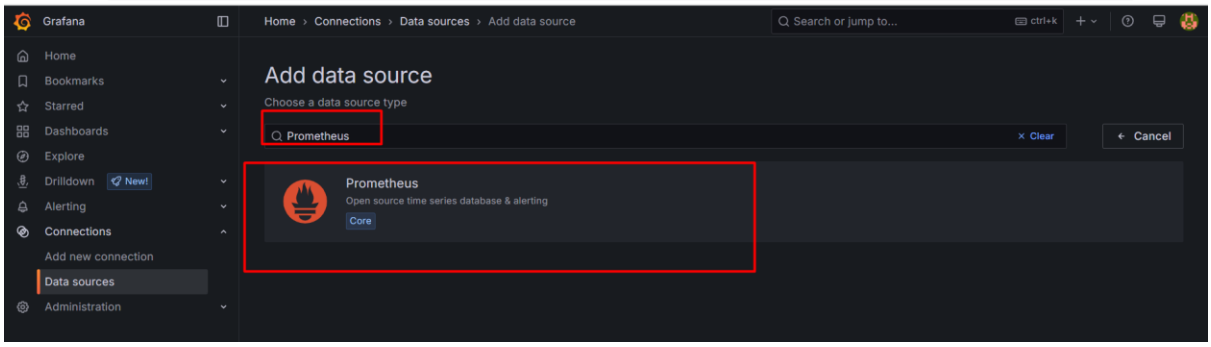
Step 7: Configure Dashboards in Grafana

1. **Log in to Grafana**.
2. Go to **Configuration → Data Sources**.



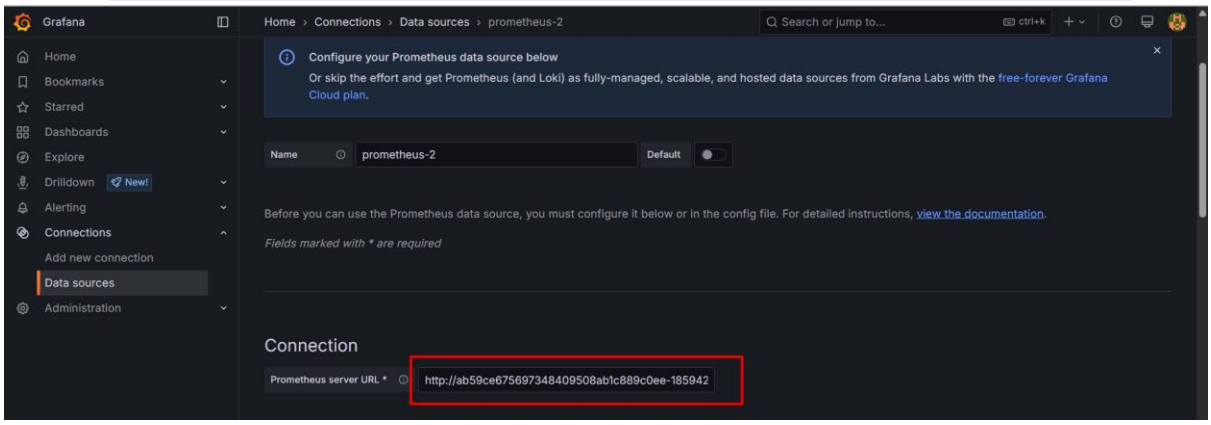


Search for Prometheus and click on

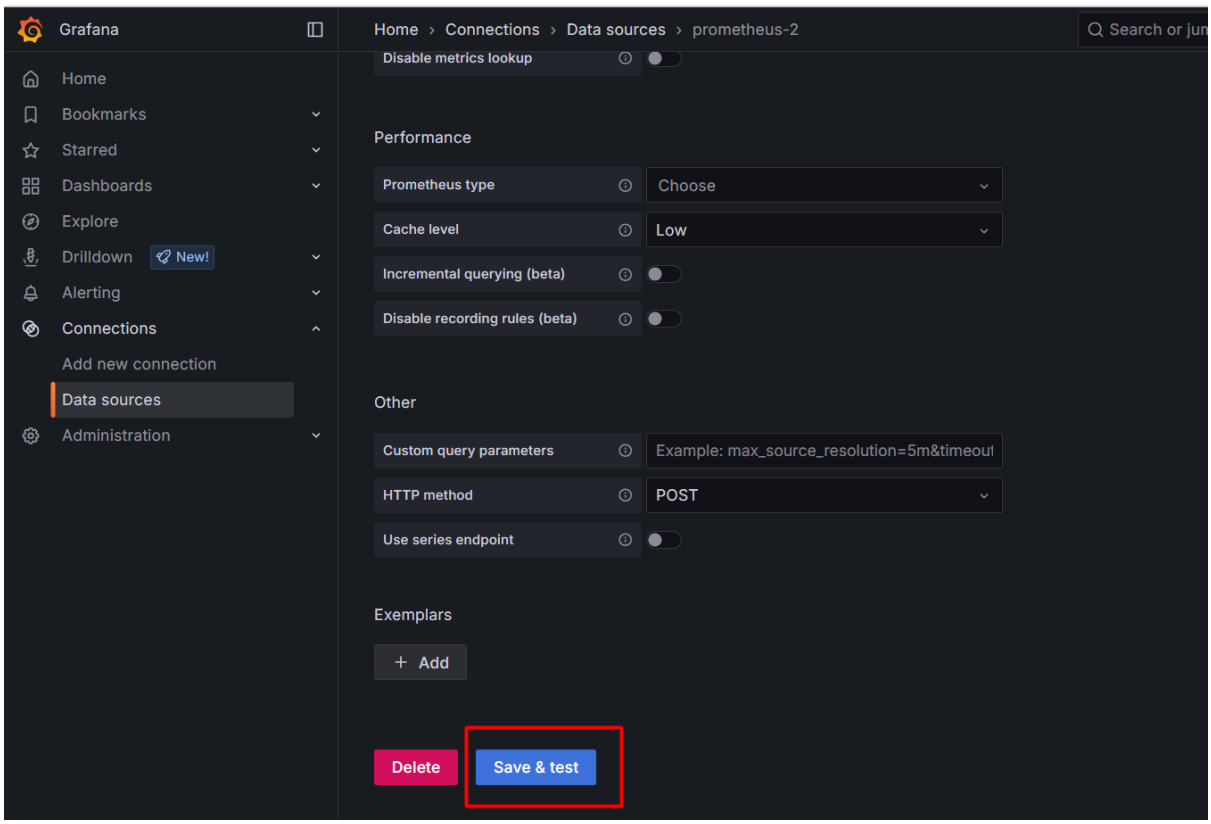


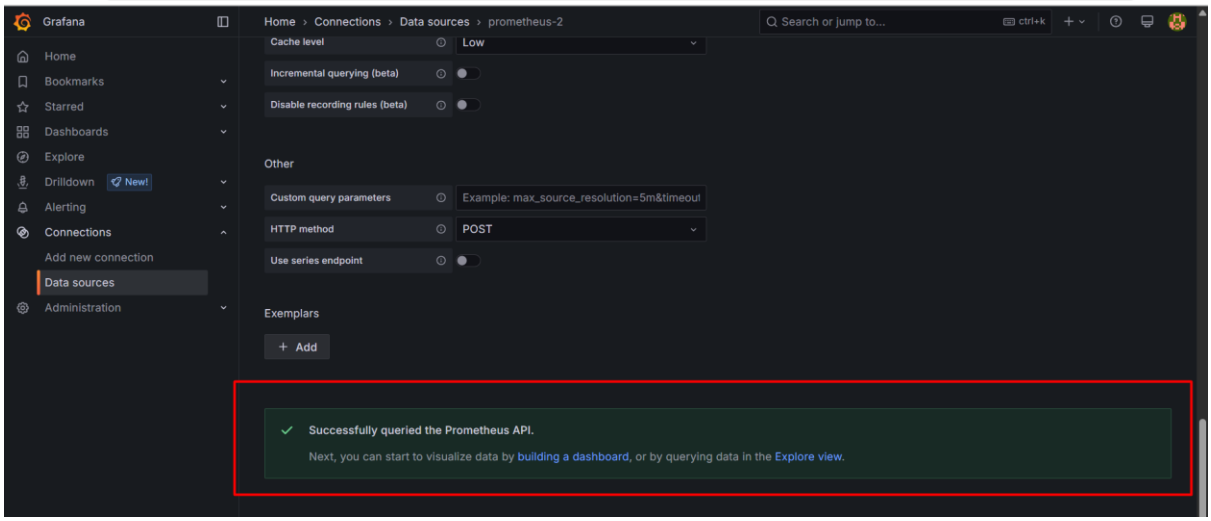
3. Add **Prometheus** as a data source:

   - **URL:** http://ab59ce675697348409508ab1c889c0ee-1859429770.sa-east-1.elb.amazonaws.com:9090

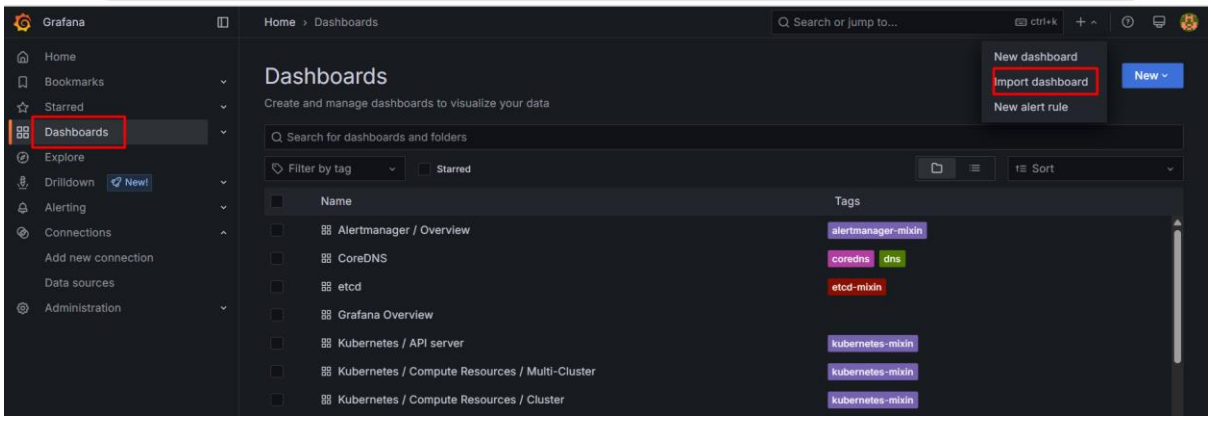


   - Click **Save & Test**.

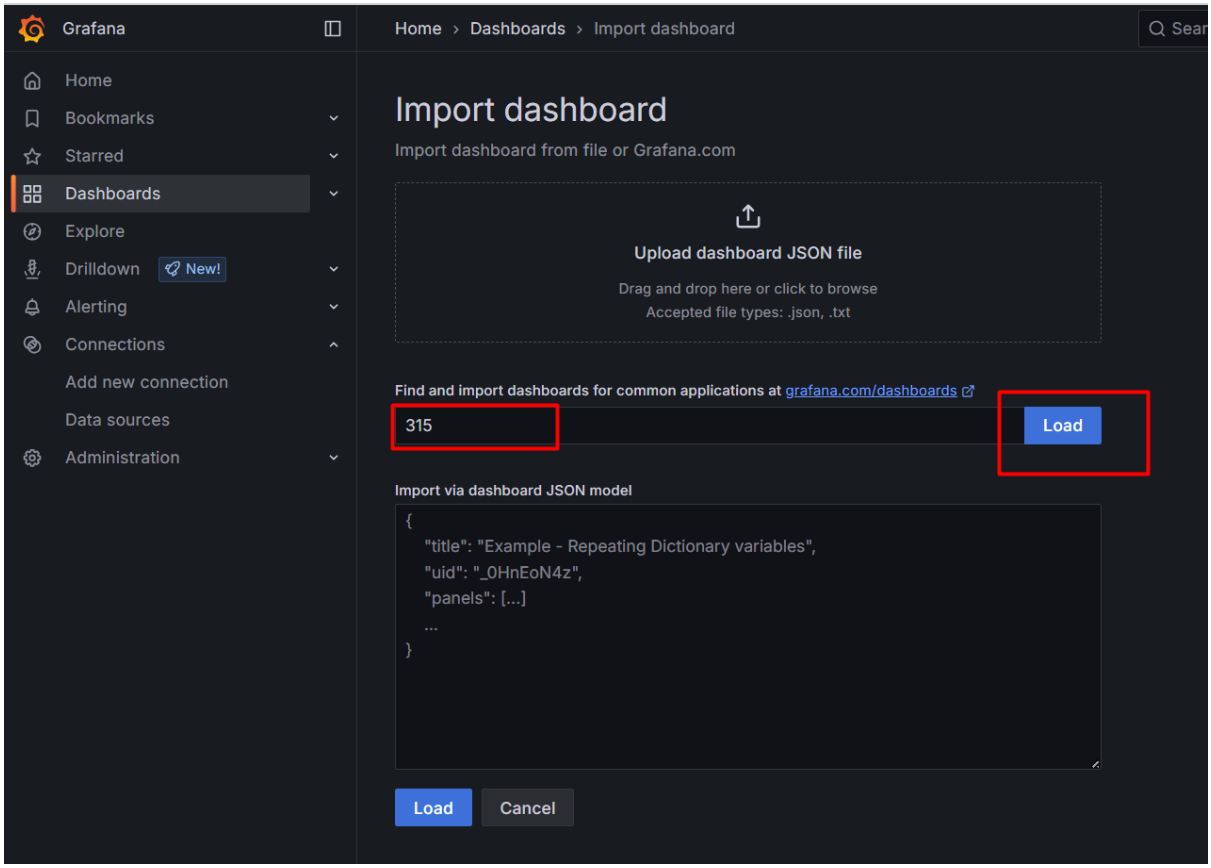

Grafana
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
New!
Alerting
Connections
Add new connection
Data sources
Administration
Home › Connections › Data sources › prometheus-2
Search or jun
Disable metrics lookup
Performance
Prometheus type
Choose
Cache level
Low
Incremental querying (beta)
Disable recording rules (beta)
Other
Custom query parameters
Example: max_source_resolution=5m&timeout
HTTP method
POST
Use series endpoint
Exemplars
+ Add
Delete
Save & test


Grafana
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
New!
Alerting
Connections
Add new connection
Data sources
Administration
Home › Connections › Data sources › prometheus-2
Search or jump to...
ctrl+k
Cache level
Low
Incremental querying (beta)
Disable recording rules (beta)
Other
Custom query parameters
Example: max_source_resolution=5m&timeout
HTTP method
POST
Use series endpoint
Exemplars
+ Add
Successfully queried the Prometheus API.
Next, you can start to visualize data by building a dashboard, or by querying data in the Explore view.

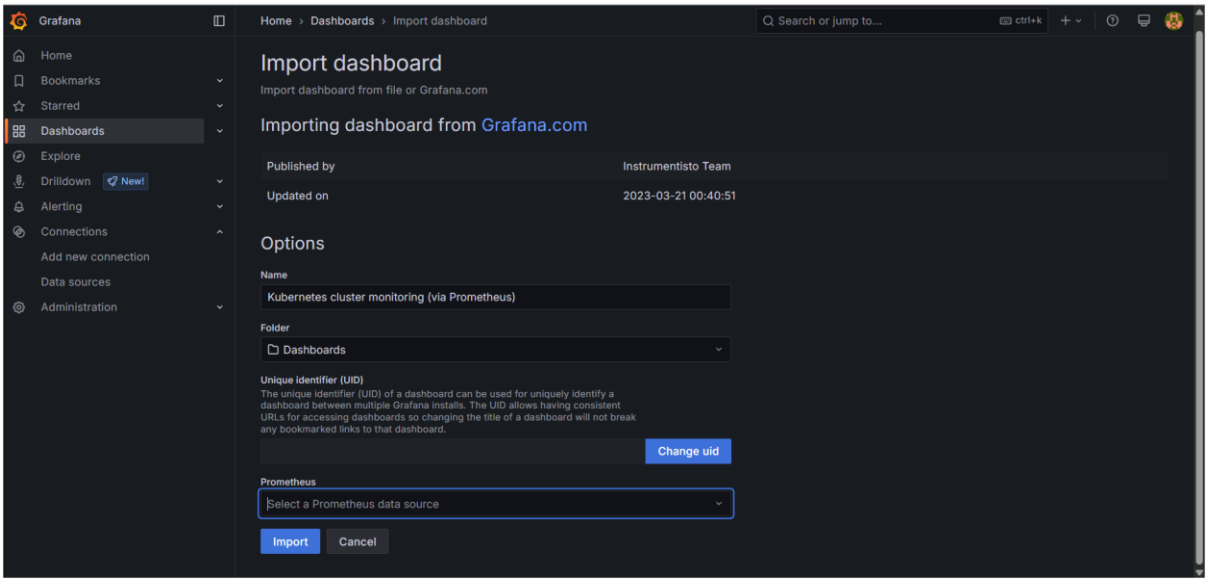
4. Go to **Dashboards → Import**, and enter **ID: 315** (Kubernetes Metrics Dashboard).



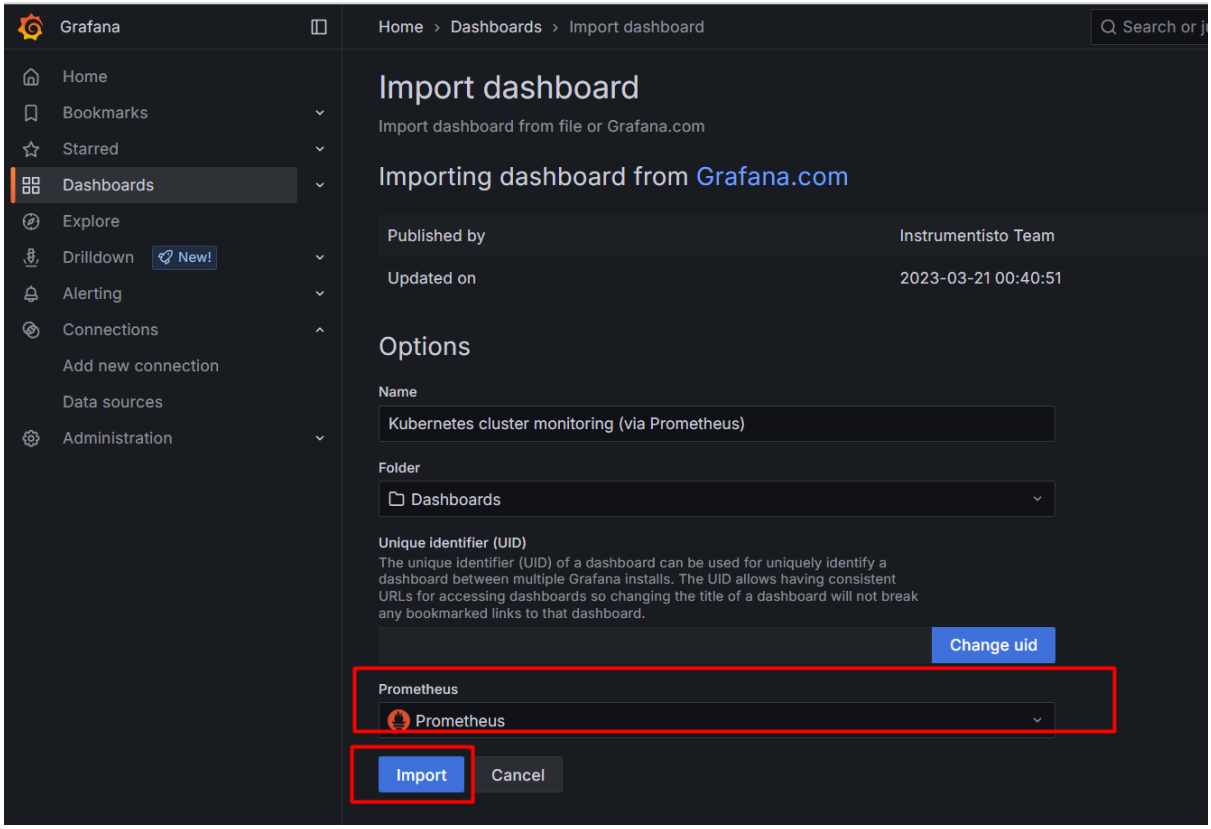
Provide k8s metrics dashboard id and click on load



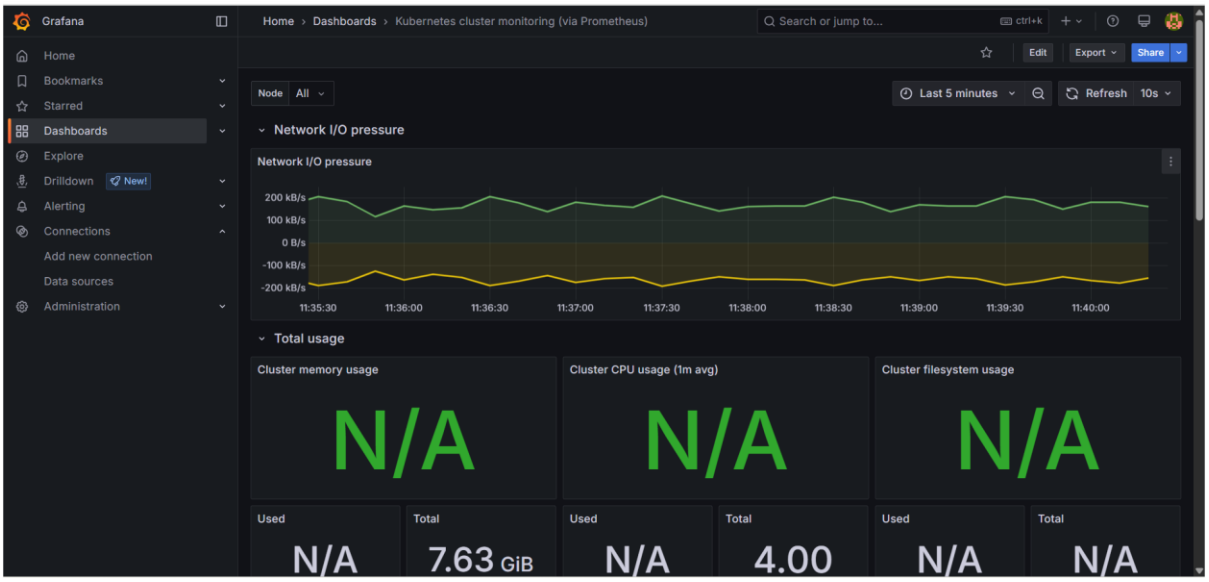
You will get page like this



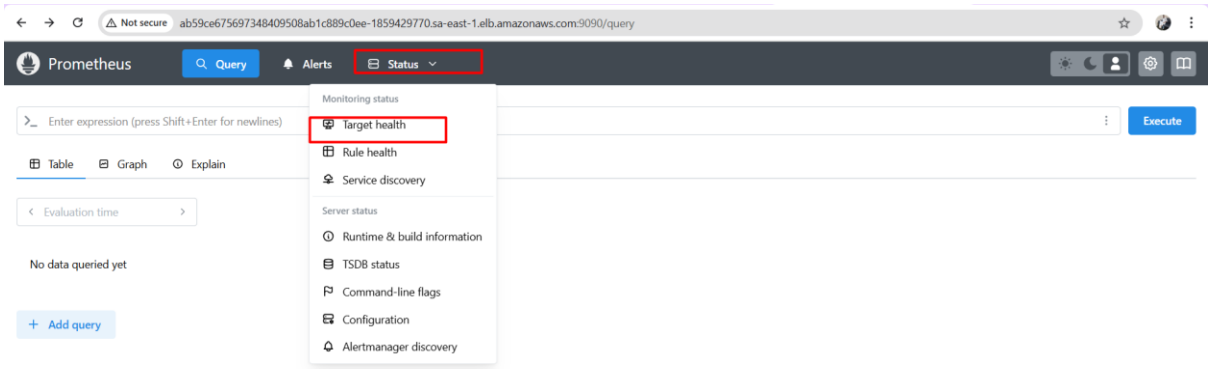
Next you need to select your Prometheus dashboard as shown below pic

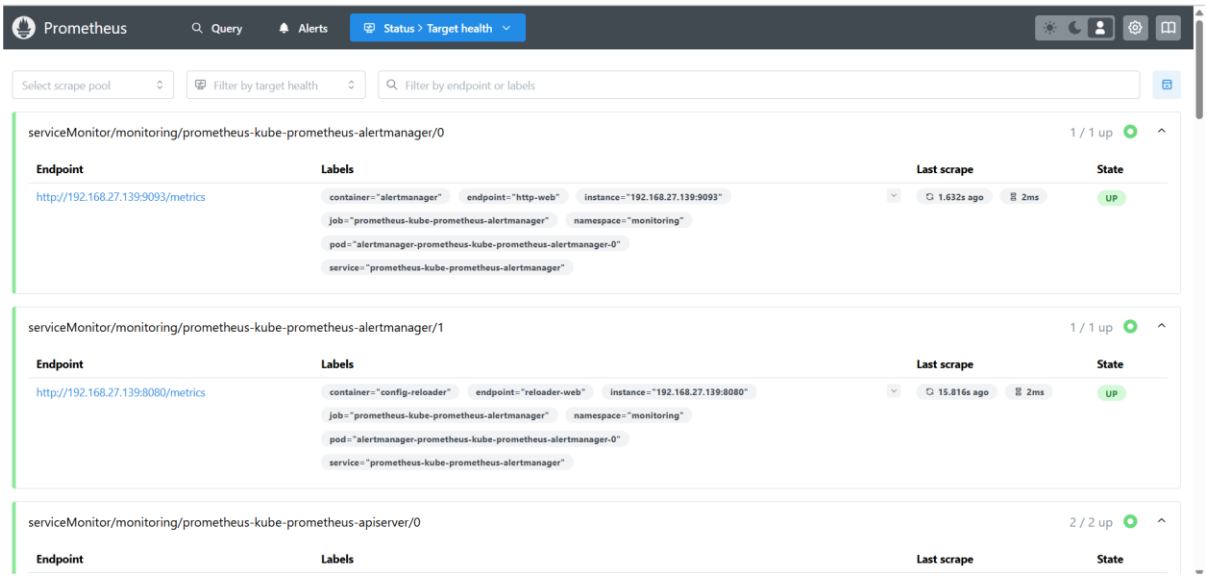


Our Grafana dashboard is ready



Now go to Prometheus and get targets





Installation commands

helm repo add prometheus-community https://prometheus-community.github.io/helm-charts

helm repo add grafana https://grafana.github.io/helm-charts

helm repo update

helm install prometheus prometheus-community/prometheus --namespace monitoring --create-namespace

helm install grafana grafana/grafana --namespace monitoring

THE END